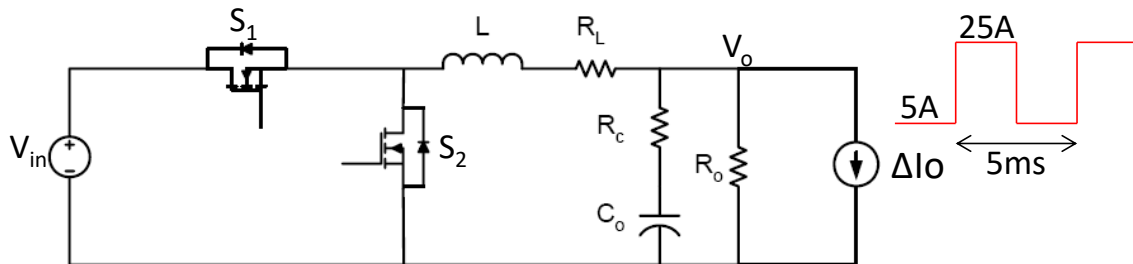


ECE 5254 - Homework #3

Buck Converter _Voltage Mode Control

Synchronous rectifier (SR) buck converter is often used in the low voltage high current applications. It uses a synchronous rectifier (MOSFET) instead of the diode in the normal buck topology. The two MOSFETs's gate signals are complementary. There is no DCM operation for the SR buck. The circuit diagram is shown below.



The parameters of the above SR buck converter are:

$V_{in}=12V$, $V_o=2V$, $L=150nH$, $R_L=2m\Omega$, $R_c=1m\Omega$, $C_o=1mF$, $R_o=200m\Omega$, $f_s=1MHz$; As shown in the graph, load transient current step $\Delta I_o=20A$ with period $T_{load}=5ms$. Load transient is assumed to be an ideal square waveform (50% duty cycle) with 0 rising and falling time. In this exercise, we assume S_1 and S_2 are ideal switches and neglect the dead-time between S_1 and S_2 .

Suggestions for simulation:

- Use the simulation file uploaded in Canvas (HW3_Ref_SR_Buck_2024.sxsch) and change the parameters.
- Make sure you put an initial voltage on the capacitor in the compensator which is in series with the resistor. (You can refer to the reference simulation file)
- Please run the transient first to find the operation point, then run AC simulation.
- Please check that, in transient simulation, output voltage is equal to what you want, and change the reference voltage to check if output voltage follows it, which ensures feedback loop works.

- 1) Using three-terminal switch model, derive open-loop output impedance transfer function. Show the bode plot and verify with SIMPLIS simulation (Frequency range: 100Hz to 1MHz). Simulate the **open-loop** load transient response in SIMPLIS (Load transient current step $\Delta I_o=20A$, 5A to 25A with period $T_{load}=5ms$), point out overshoot voltage during load step down, undershoot voltage during load step up, and settling time for both cases. (15%).

Notes:

The transfer function should be written in the analytical polynomial form as follows to get full points:

$$Z_{out}(s) = Z_0 \frac{1 + N_1s + N_2s^2 + N_3s^3 + \dots}{1 + D_1s + D_2s^2 + D_3s^3 + \dots}$$

You need to use the symbols of the above given circuit parameters (V_{in} , V_o , ..., f_s) to derive the expression. Please plot the model result and SIMPLIS on the same bode plot.

2) Close the loop with voltage mode control: (50%)

- a. Design the type III compensator to achieve 80kHz bandwidth (your bandwidth should be within $\pm 5\text{kHz}$ window, i.e, 75kHz to 85kHz) with at least 60 degree phase margin and at least 10dB gain margin (it is OK if your gain margin is infinite, i.e, your phase never reaches -180 degree.). Show your critical design steps and show the values of your resistors and capacitors in your final compensator. Please use some practical values in the compensator design. For resistors, 10Ω to $1\text{M}\Omega$ and for capacitors, 1pF to $100\mu\text{F}$ are fine. (20%).
- b. Show bode plot of your designed compensator transfer function and verify the transfer function with SIMPLIS simulation. Show bode plot of your designed loop gain transfer function and verify with SIMPLIS simulation (Mark the phase margin and gain margin). Show the bode plot of closed-loop output impedance and verify with SIMPLIS simulation. Please plot the model results and SIMPLIS on the same bode plot. (10%)
- c. Simulate the same load step transient response in SIMPLIS as Question (1) and compare the results (overshoot, undershoot voltage and settling time) with open-loop case. Comment on the comparisons. (10%)
- d. Change the first zero of your compensator design to 1/5 of the previous value while keeping the loop gain bandwidth still at 80kHz and at least 60 degree phase margin and at least 10dB gain margin. Please show bode plots of simulated loop gain both for this case and previous case in one graph. Compare the load transient performance with previous

case. State your observations and explain the reason behind the observations. (10%)

3) Different loop gain bandwidth design: (35%)

- a. Define the previous case $f_c=80\text{kHz}$ as case 1, now change loop gain bandwidth as the following two cases: Case 2: $f_c=1\text{kHz}$ with at least 60 degree phase margin and at least 10dB gain margin; Case 3: $f_c=200\text{kHz}$ with at least 60 degree phase margin and at least 10dB gain margin. Describe what you change to increase and decrease the bandwidth. Show the bode plots of simulated loop gain for three cases in one graph (Mark the phase margin and gain margin for each case). You do not need to provide the detailed design steps for this part. (15%)
- b. Show the closed-loop output impedances for these three cases in one graph and show the load step transient responses (overshoot, undershoot, and settling time) for these three cases in another graph. State your observations from these two graphs and explain the reason behind the observations. (10%)
- c. **Discussion:** the overshoot and undershoot voltage are very important specs in many applications, does continuously pushing bandwidth to higher frequency keep reducing the overshoot and undershoot voltage? State your opinion and explain the reason. (10%)

(Hint: In SIMPLIS simulation, you can try to further push bandwidth to Case 4: $f_c=300\text{kHz}$ in the above example, then compare the overshoot and undershoot performances between Case 1: $f_c=80\text{kHz}$, Case 3: $f_c=200\text{kHz}$ and Case 4: $f_c=300\text{kHz}$).

Due time: Nov. 8th, 2024. 11:30pm. All the homework should be submitted into canvas before due time. Late homework should be sent into TA's email sundaramoorthy@vt.edu (10% penalty per day after due time will be strictly executed unless prior arrangements have been made with the instructor). For the report, pdf file is accepted. Please change your file name to HW#_First name_Last name for easy recognition. Please put your name on the first (or cover) page of your report. Computer typed report will be appreciated.

All the work MUST be your own; the VT honor code will be enforced.